

Project - Petar Matejčić

Petar Matejčić

September 24, 2026

1 Introduction

Bayesian additive regression trees (BART) is a nonparametric Bayesian regression approach based on a sum-of-trees model where fitting and inference are done using a Bayesian backfitting MCMC algorithm. The sum-of-trees approach affords BART flexibility to represent nonlinear effects and interactions between variables. It also allows for various feature selection schemes using the splitting rules in the posterior tree samples.

The authors in the original BART paper proposed using variable usage frequencies (or variable inclusion proportions) in the posterior trees for feature selection. This offers a simple importance measure based on the intuition that an important variable will appear more frequently as a splitting rule in a regression tree. Bleich et al. then expanded on this approach with a permutation-based thresholding scheme.

A key property of variable inclusion proportions (VIPs) is that every appearance of a variable contributes equally to the overall importance score. This means that VIPs measure only how often a predictor is used, not how much it adds to the tree structure. The log-marginal-likelihood-weighted approach therefore weights each appearance of a variable in an internal node with the positive log-marginal-likelihood-gain associated with the subtree rooted at that node.

The goal of this report is to investigate if a log-marginal-likelihood-weighted importance measure improves feature selection compared to standard variable inclusion proportions. To accomplish this we use six data-generating scenarios which allow us to know which variables are truly relevant. Additionally, the two feature selection methods were compared with regard to sensitivity to

BART hyperparameter settings. We show that in almost all tested scenarios the weighted method achieved better performance statistics than standard VIPs. We also demonstrate how those results are robust to changes in tested BART hyperparameters.

In Section 2 we briefly introduce BART, standard variable inclusion proportions and the permutation-based thresholding scheme. In Section 3 we introduce the log-marginal-likelihood-weighted method in detail. In Section 4 we present results of the simulation experiments and the sensitivity analysis.

2 BART

Bayesian additive regression trees (BART) is a Bayesian sum-of-trees approach to the problem of making inference about an unknown function f which predicts an output Y from an input vector x :

$$Y = f(x) + e$$

where x is a p dimensional vector and $e \sim N(0, \sigma^2)$.

In other words, we are trying to model or approximate $f(x) = \mathbb{E}(Y|x)$ by a sum of m regression trees

$$f(x) \approx h(x) = \sum_{i=1}^m g_i(x; T_i, M_i)$$

$$Y \approx \sum_{i=1}^m g_i(x; T_i, M_i) + \epsilon$$

Each regression tree is defined by two components: a tree structure T_i defined by splitting rules assigned to internal nodes and a set of terminal node parameters $M_i = \{\mu_1, \dots, \mu_b\}$. The splitting rules are binary splits of the form $\{x \in A\}$ vs. $\{x \notin A\}$, usually dependent on a single component of $x = (x_1, \dots, x_p)$: for continuous variables they are typically of the form $x_j \leq c$ vs. $x_j > c$, while for categorical variables A is usually a subset of values variable x_j can attain. Evaluation $g_i(x; T_i, M_i)$ assigns the value μ_i of a terminal node arrived at by the sequence of splitting rules followed from the root of the tree. Thus, a sum-of-trees model can be described by the set of all parameters $(T_1, M_1), \dots, (T_m, M_m)$ assigned to the regression trees,

together with the global parameter σ .

The Bayesian approach comes from imposing a prior over all the unknown parameters in the model and sampling from the posterior distribution over the sum-of-trees model space using a backfitting algorithm. It is worth pointing out that the prior distributions are set to give preference to small, simple trees with small effects. This results in a flexible model where each tree is a “weak learner” and no tree dominates the result, while both interactions (via trees dependent on multiple variables) and additive effects (via trees dependent on only one variable) can be represented.

Another important consideration is that the priors are chosen such that they satisfy:

$$\begin{aligned} p((T_1, M_1), \dots, (T_m, M_m), \sigma) &= p(\sigma) \prod_j p(T_j, M_j) \\ &= p(\sigma) \prod_j p(M_j|T_j)p(T_j) \end{aligned}$$

and:

$$p(M_j|T_j) = \prod_i p(\mu_{ij}|T_j).$$

In other words, under these assumptions σ is independent of tree parameters, tree parameters (T_j, M_j) are independent of each other and terminal node parameters M_j of each tree are mutually independent, conditionally on tree structure T_j . This greatly simplifies the problem of prior specification as only $p(T_j)$, $p(\mu_{ij}|T_j)$ and $p(\sigma)$ are needed.

Finally, using a Bayesian backfitting MCMC algorithm, after an appropriate burn-in, a sequence of draws $f_1^*, \dots, f_K^*$, can be taken as an approximate, dependent sample of size K from $p(f|y)$. This sample can then be used for posterior inference. For example, the mean of the sample can be used to approximate the posterior mean $\mathbb{E}(f(x)|y)$, or, as we will show, the sample can be used in feature selection.

2.1 Variable Inclusion Proportions

Chipman, George and McCulloch in [1] described a simple method for feature selection using variable inclusion proportions. This approach is based on the idea that relevant variables should be more often used in splitting rules in

the sum-of-trees models than irrelevant variables.

Formally, for each sampled sum-of-trees model f_k^* let z_{ik} be the proportion of all splitting rules in the model which use the i th variable. We then estimate the variable inclusion proportion of the i th variable as the posterior mean over all posterior samples:

$$v_i := \frac{1}{K} \sum_{k=1}^K z_{ik}$$

Since VIPs must sum to one, this is a measure of relative importance, meaning if one variable has a higher variable inclusion proportion, others necessarily have smaller inclusion proportions. Once we have calculated the VIPs, we can rank variables by their estimated VIPs, selecting those with “high-enough” proportions. In section 2.2 we briefly go over an approach by Bleich et al. for determining what constitutes “high-enough”.

It is worth pointing out a few details about VIPs. First of all, it appears that smaller values of m are more suitable for feature selection, because for larger m , the number of trees allows for the use of irrelevant variables without compromising the predictive performance. Smaller m , on the other hand, in effect create a bottleneck, forcing the variables to compete and thereby tending to exclude irrelevant variables.

Secondly, it is important to note that VIPs measure variable usage frequency across posterior sampled sum-of-trees models, not their contribution to model fit. Chipman, George and McCulloch note that it might be useful to weight variables by the number of data points present in the node. In section 3.2 we will introduce one similar weighting method based on log-marginal likelihoods.

2.2 Variable Selection Using VIPs

We now turn to the question of feature selection based on VIPs. Let us first briefly discuss a “null” data setting where we have p predictor variables unrelated to the outcome variable Y . In this setting we might expect splitting rules to use predictor variables uniformly, with the result that $v_i \approx 1/p$ for all i .

Bleich et al. in [2] demonstrated empirically that this is not always the case. They hypothesize this is due to two factors: firstly, even though the

predictors and the response are independent, there will naturally be some random association between them in the dataset and BART could take advantage of that. Secondly, the issue might stem from the BART procedure, possibly due to the Gibbs sampler in the MCMC backfitting algorithm hitting a local maximum, as the authors note.

Thus, Bleich et al. in [2] devised a scheme for feature selection using permutation-based null importance. Specifically, they suggest creating P permutations of the response variable in the dataset, fitting a BART model for each permutation and, finally, calculating the VIPs for each permutation. Notice that by permuting only the response, we remove any dependency between the predictors and the response, but we maintain the possible interdependencies between various predictor variables. Thus, for each variable, we generate P variable inclusion proportions, in effect obtaining a null distribution under the hypothesis that the response is unrelated to the predictors.

Finally, Bleich et al. describe three methods for selecting a threshold for feature selection:

1. *local* - for each variable we select a threshold based on the $1 - \alpha$ quantile of its permutation null distribution
2. *global-max* - for each permutation p we take the maximum VIP obtained by all variables v_p^* ; we then select a constant threshold for all variables as the $1 - \alpha$ quantile of all the maxima $v_1^*, \dots, v_P^*$
3. *global-SE* - we calculate a global multiplier C^* based on means m_k and standard deviations s_k of the null distributions; we then use $m_k + C^* s_k$ as the threshold for the k th variable

For more details on each threshold see [2]. Here it is worth noting that the first strategy is the least strict, as the variable inclusion proportion for a variable needs to only exceed extreme values in its own null distribution. The global-max method, on the other hand, is the strictest, as a variable's VIP needs to be extreme relative to all other null distributions. The third method, global-SE, lies somewhere in the middle: it retains some of the strictness of global-max by using a global multiplier, but it takes into account the null distributions of each variable.

While this procedure was developed for the standard variable inclusion proportions described by Chipman et al., it can easily be adapted for weighted VIPs

because the thresholding procedure stays the same and only the definition of importance changes.

3 Log-Marginal Likelihood Importance

As discussed above, variable inclusion proportions measure only how often a given variable is used in a splitting rule. This motivates us to develop a weighting method which will add more information about the tree structure to the importance statistic. To achieve this we use marginal likelihood to compare a given subtree with a simpler alternative. In Subsection 3.1 we introduce marginal likelihood; in Subsection 3.2 we formally define the weighted statistic and discuss its properties; finally, in Subsection 3.3 we describe how the thresholding scheme proposed by Bleich et al. can be used with the weighted statistic.

3.1 Marginal Likelihood

We now go into more details on the marginal likelihood of a regression tree. Let us denote $T_{(j)}$ as the set of all trees in a sum-of-trees model except T_j and $M_{(j)}$ as the appropriate terminal node parameters.

Firstly, we will consider the conditional probability

$$p(T_j, M_j | T_{(j)}, M_{(j)}, \sigma, Y).$$

Notice how the dependence on $(T_{(j)}, M_{(j)}, Y)$ can be condensed down to dependence on partial residuals:

$$R_j := Y - \sum_{k \neq j} g(x; T_k, M_k)$$

Therefore, the above conditional probability is equivalent to:

$$p(T_j, M_j | R_j, \sigma).$$

As we are interested only in how well the regression tree structure explains the data, we will treat M_j as nuisance parameters. In other words we will average over all terminal node parameters by integrating over them. Thus,

using Bayes rule we obtain the equation:

$$p(T_j|R_j, \sigma) \propto p(T_j) \int p(R_j|M_j, T_j, \sigma)p(M_j|T_j, \sigma)dM_j.$$

We call the integral on the right-hand side *the marginal likelihood of the regression tree T_j* . This integral can be evaluated analytically in closed form because BART uses a conjugate prior for terminal node parameters M_j . For more details see [1].

It is worth breaking down the calculation of the marginal likelihood on an example. Suppose a regression tree T_j has b terminal nodes with parameters $M_j = \{\mu_{1j}, \dots, \mu_{bj}\}$. Since each partial residual falls into only one terminal node it follows:

$$p(R_j|M_j, T_j, \sigma) = \prod_{i=1}^b p(R_j|\mu_{ij}, \sigma).$$

Furthermore, because BART assumes that $\mu_{1j}, \dots, \mu_{bj}$ are conditionally independent given T_j , we have:

$$p(M_j|T_j, \sigma) = \prod_{i=1}^b p(\mu_{ij}|T_j, \sigma).$$

Combining the two and factoring the integrals we obtain:

$$\begin{aligned} p(R_j|T_j, \sigma) &= \int \left[\prod_{i=1}^b p(R_j|\mu_{ij}, \sigma)p(\mu_{ij}|T_j, \sigma) \right] dM_j \\ &= \prod_{i=1}^b \left[\int p(R_j|\mu_{ij}, \sigma)p(\mu_{ij}|T_j, \sigma)d\mu_{ij} \right] \end{aligned}$$

Therefore, calculating the marginal likelihood of regression tree T_j comes down to evaluating one integral for each terminal node. For numerical stability and convenience, it is useful to work with the equivalent *log-marginal likelihood* which converts the product over terminal nodes into a sum.

3.2 Log-Marginal Likelihood Weighting

We will now formally introduce an alternative feature selection scheme based on variable inclusion proportions discussed in section 2.1. In this scheme we will still look at the proportion of times a variable is used for a splitting rule within a sum-of-trees model, but this time each appearance will be weighted by a log-marginal likelihood gain.

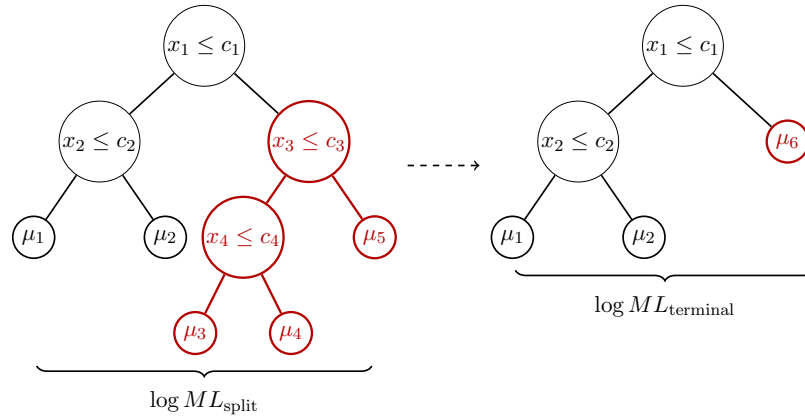


Figure 1. Illustration of the subtree-collapsing step used to calculate the weight of a splitting rule. The left tree shows the original configuration used to compute $\log ML_{\text{split}}$. In the right tree, the subtree rooted at the selected internal node is replaced by a single terminal node, giving $\log ML_{\text{terminal}}$. The difference between these two log marginal likelihoods determines the weight assigned to the splitting rule.

In particular, for a fixed tree in the model and a fixed splitting rule (or, in other words, a fixed internal node v), we will consider an alternative tree where that internal node v (and the subtree following it) is replaced by a terminal node. We then calculate the log-marginal likelihood $\log ML_{\text{split}}$ of the original configuration and log-marginal likelihood of the collapsed configuration $\log ML_{\text{terminal}}$. Finally, we define weights w_v as:

$$\begin{aligned}\delta_v &:= \log ML_{\text{split}} - \log ML_{\text{terminal}} \\ w_v &:= \max(\delta_v, 0)\end{aligned}$$

Thus, for each sampled sum-of-trees model f_k^* we consider $\tilde{z}_{ik}$ the weighted

proportion of all splitting rules in the model which use the i th variable:

$$\tilde{z}_{ik} := \frac{\sum_{v \in V_k} \mathbb{1}(s(v) = i) w_v}{\sum_{v \in V_k} w_v},$$

where V_k is the set of all internal nodes in f_k^* and $s(v)$ is the variable used in the splitting rule assigned to v .

We average the proportions across all posterior draws $f_1^*, \dots, f_K^*$, obtaining the final importance statistics:

$$\tilde{v}_i := \frac{1}{K} \sum_{k=1}^K \tilde{z}_{ik}.$$

Before we move on to experimental results, it is worth pointing out some details about this metric.

Firstly, let us go into more depth on interpretation of the importance statistic. A large w_v means that, for observations reaching v , the original subtree has a larger marginal likelihood than treating the whole region as a terminal node. Thus, we can say that the original subtree structure is more strongly supported by the partial residuals than the alternative simpler tree. This is still a relative importance measure as we are normalizing by the total weight in the posterior sample, therefore we should interpret the scores as shares of the weighted splitting evidence. We truncate the weights at 0 in order to preserve this proportional interpretation. The weights then represent positive evidence for retaining the subtree rooted at node v .

Secondly, it is important to note that weights between nodes somewhat overlap. As we replace the whole subtree stemming from node v with a single node some of the marginal likelihood gain will come from subsequent nodes in the subtree and not from v itself. While this is not necessarily undesirable (the split at v might be important to the following splitting rules), it is a trade-off, as variables used near the root of the tree may tend to have higher weights. BART regularization may limit the magnitude of the issue by favoring smaller trees, but the overlapping attribution remains a property of the statistic.

Lastly, since log-marginal likelihood is already used as part of the BART backfitting MCMC algorithm its implementation can be reused for the purpose of feature selection. Additionally, as discussed at the end of section 3,

log-marginal likelihood can be evaluated as a sum of closed-form solutions to the terminal node integrals. Notice that in the expression for δ_v terminal nodes outside the subtree rooted at v cancel out. Therefore, for $\log \text{ML}_{\text{split}}$ only summands associated with terminal nodes in the subtree need to be evaluated. Likewise, for $\log \text{ML}_{\text{terminal}}$ only a single terminal node contribution needs to be evaluated.

The implementation of the log-ML-weighted importance statistic is available in the accompanying BART library at github.com/PetarMatejcic/GenBART.

3.3 Feature Selection with Weighted VIPs

Because this is a relative importance measure structurally similar to variable inclusion proportions, we can use the feature selection scheme proposed by Bleich et al. We simply replace VIPs with the new log-marginal-likelihood-weighted importance statistics. Therefore the underlying procedure remains the same for both statistics: a permutation procedure is used to obtain the null distributions and the same thresholding rules apply to both statistics.

This is important when interpreting the experimental results. Because the BART fitting procedure, permutation scheme and thresholding rules are held fixed, the differences in feature selection results can be attributed to the choice of importance statistics, allowing for Monte Carlo variation.

4 Simulation Experiments

We can now present results comparing log-ML-weighted VIPs with standard variable inclusion proportions. In the first set of experiments we assessed variable recovery performance of both methods under six different data-generating scenarios. In the second set, we re-ran three scenarios under different BART hyperparameter settings to determine whether the results of the previous experiments were robust to the tested hyperparameters. Subsections 4.1 - 4.4 present results of experiments comparing performance of the two methods, while Subsection 4.5 presents the BART hyperparameter sensitivity analysis.

4.1 General Setup

In the following experiments we were interested in *whether the log-ML-weighted importance statistic recovers relevant predictors better than raw VIPs* and *how the two methods behave when predictors are correlated or there is significant noise in the data*.

To compare the two importance statistics we use synthetic datasets taken from [3]. Using generated datasets is convenient because the set of truly relevant predictors is known and correlation and noise can be controlled. These data-generating scenarios cover cases of both continuous and binary response variables, continuous and mixed (binary and continuous) predictor variables, as well as different degrees of correlation between predictors and noise levels.

An additional consideration was how the number of irrelevant predictors affects feature selection. Therefore, for a fixed number of relevant predictors in a scenario, varying numbers of additional irrelevant variables were added, thus increasing the dimensionality p of the data. Similarly, several sample sizes n were used for each scenario to see whether the methods become more reliable with more data. Each scenario was replicated 100 times and relevant statistics were recorded.

Several statistics were used for comparison: *precision* (proportion of true relevant variables among the selected variables), *recall* (proportion of truly relevant variables detected), *F1-score* and r_{miss} (proportion of replications missing at least one relevant variable). *Precision*, *recall* and *F1-score* statistics in the following tables are averages over 100 replications.

Within each replication the generated datasets are held fixed, thus both methods work on the same dataset. Similarly, BART specifications, permutation scheme and thresholding rules are kept the same for both importance statistics. The *global-SE* thresholding scheme was used in all experiments. Code used to reproduce the simulation experiments and hyperparameter sensitivity analysis is available at github.com/PetarMatejcic/BART_logml.vi.

4.2 Continuous Predictors and Continuous Response

We start with two scenarios with continuous predictor and continuous response variables. For both scenarios values $n = 500$, $p \in \{50, 200\}$ and $\sigma^2 = 1$ were considered.

Scenario 1. Predictor variables $x = (x_1, \dots, x_p)$ are sampled independently from a Uniform(0, 1) distribution. The response y is generated according to $y = f(x) + \epsilon$, where $\epsilon \sim N(0, \sigma^2)$ and f is defined as:

$$f(x) = 10 \sin(\pi x_1 x_2) + 20(x_3 - 0.5)^2 + 10x_4 + 5x_5.$$

Table 1. Variable selection results for Scenario 1. The upper panel reports mean performance statistics across 100 replications, with standard errors in parentheses. For each performance metric, the better value within a configuration is shown in bold. The lower panel reports the mean number of selected variables, the proportions of replications selecting exactly, fewer than, or more than the true number of relevant variables, and the mean excess number of selected variables.

Selection performance							
n	p	σ^2	Importance	Precision	Recall	F1	r_{miss}
500	50	1	Raw	0.951 (0.01)	0.904 (0.01)	0.921 (0.01)	40%
			LogML	1.000 (0.00)	0.844 (0.01)	0.911 (0.01)	69%
500	200	1	Raw	0.434 (0.01)	0.718 (0.02)	0.531 (0.01)	88%
			LogML	0.642 (0.02)	0.966 (0.01)	0.762 (0.01)	17%

Selection size diagnostics (true = 5)								
n	p	σ^2	Importance	Mean	Exact	Under	Over	Mean excess
500	50	1	Raw	4.78	52%	33%	15%	0.16
			LogML	4.22	31%	69%	0%	0.00
500	200	1	Raw	8.71	8%	1%	91%	3.72
			LogML	7.90	6%	1%	93%	2.91

Scenario 2. Predictor variables $x = (x_1, \dots, x_p)$ are generated from a multivariate normal distribution $N(0, \Sigma)$, where Σ is defined component-wise with $\Sigma_{ij} = 0.3^{|i-j|}$. The response y is then generated according to $y = f(x) + \epsilon$, where $\epsilon \sim N(0, \sigma^2)$ and f is defined as:

$$f(x) = 2x_1x_4 + 2x_7x_{10}.$$

Table 2. Variable selection results for Scenario 2. Statistics and notation are as in Table 1.

Selection performance							
n	p	σ^2	Importance	Precision	Recall	F1	r_{miss}
500	50	1	Raw	0.200 (0.03)	0.113 (0.02)	0.138 (0.02)	100%
			LogML	0.327 (0.03)	0.240 (0.02)	0.267 (0.02)	100%
500	200	1	Raw	0.040 (0.01)	0.075 (0.01)	0.051 (0.01)	100%
			LogML	0.068 (0.01)	0.212 (0.02)	0.102 (0.01)	100%

Selection size diagnostics (true = 4)								
n	p	σ^2	Importance	Mean	Exact	Under	Over	Mean excess
500	50	1	Raw	4.78	28%	5%	67%	0.83
			LogML	4.22	60%	9%	31%	0.31
500	200	1	Raw	8.71	1%	0%	99%	4.71
			LogML	7.90	1%	0%	99%	3.90

As can be seen from tables 1 and 2, the log-ML-weighted importance statistic achieves better results in most continuous-response experiments.

The only experiment where standard VIPs produced better results was Scenario 1 with $p = 50$ variables. Here the log-ML-weighted method performed markedly worse, selecting fewer than the true number of relevant features in 69% of the repeats. However, in the same scenario with $p = 200$ variables, the log-ML-weighted statistic performed better with regard to all statistics. In particular the percentage of repeats with a missed truly important variable was only 17% compared with 88% for standard VIPs while the number of selected features was on average lower (7.9 compared to 8.7).

In Scenario 2 both methods performed relatively poorly, attesting, perhaps, to the difficulty of the problem. The log-ML-weighted method yields higher *precision*, *recall* and *F1-score*, while r_{miss} was 100% for both methods meaning neither recovered the complete relevant-variable set in any replication.

4.3 Mixed Predictors and Continuous Response

In the following two scenarios some of the predictor variables are sampled from continuous distributions, while the rest are sampled from discrete distributions.

Scenario 3. Predictor variables $x_1, \dots, x_{\lceil p/2 \rceil}$ are sampled independently from a Bernoulli(0.5) distribution, while predictors $x_{\lceil p/2 \rceil+1}, \dots, x_p$ are sampled independently from a Uniform(0, 1) distribution. The response y is generated according to $y = f(x) + \epsilon$, where $\epsilon \sim N(0, \sigma^2)$ and f is defined as:

$$f(x) = 10 \sin(\pi x_{\lceil p/2 \rceil+1} x_{\lceil p/2 \rceil+2}) + 20(x_{\lceil p/2 \rceil+3} - 0.5)^2 + 10x_1 + 5x_2.$$

Feature selection methods were compared for combinations of values $n \in \{500, 1000\}$, $p \in \{50, 200\}$ and $\sigma^2 \in \{1, 10\}$.

Table 3. Variable selection results for Scenario 3. Statistics and notation are as in Table 1.

Selection performance							
n	p	σ^2	Importance	Precision	Recall	F1	r_{miss}
500	50	1	Raw	0.929 (0.01)	0.594 (0.01)	0.718 (0.01)	100%
			LogML	1.000 (0.00)	0.808 (0.01)	0.891 (0.01)	88%
500	50	10	Raw	0.887 (0.01)	0.562 (0.01)	0.680 (0.01)	100%
			LogML	1.000 (0.00)	0.786 (0.01)	0.877 (0.01)	95%
500	200	1	Raw	0.359 (0.01)	0.512 (0.01)	0.412 (0.01)	100%
			LogML	0.603 (0.01)	0.904 (0.01)	0.713 (0.01)	46%
500	200	10	Raw	0.299 (0.01)	0.454 (0.01)	0.352 (0.01)	100%
			LogML	0.547 (0.01)	0.880 (0.01)	0.666 (0.01)	60%
1000	50	1	Raw	0.944 (0.01)	0.618 (0.01)	0.743 (0.01)	100%
			LogML	1.000 (0.00)	0.818 (0.01)	0.897 (0.01)	84%
1000	50	10	Raw	0.933 (0.01)	0.600 (0.01)	0.725 (0.01)	100%
			LogML	0.998 (0.00)	0.804 (0.01)	0.886 (0.01)	83%
1000	200	1	Raw	0.378 (0.01)	0.596 (0.01)	0.454 (0.01)	100%
			LogML	0.713 (0.01)	0.950 (0.01)	0.807 (0.01)	24%
1000	200	10	Raw	0.372 (0.01)	0.562 (0.01)	0.439 (0.01)	100%
			LogML	0.668 (0.01)	0.946 (0.01)	0.773 (0.01)	26%

Selection size diagnostics (true = 5)								
n	p	σ^2	Importance	Mean	Exact	Under	Over	Mean excess
500	50	1	Raw	3.27	4%	95%	1%	0.01
			LogML	4.04	12%	88%	0%	0.00
500	50	10	Raw	3.26	4%	95%	1%	0.01
			LogML	3.93	5%	95%	0%	0.00
500	200	1	Raw	7.61	11%	7%	82%	2.70
			LogML	7.89	7%	3%	90%	2.92
500	200	10	Raw	8.23	7%	4%	89%	3.29
			LogML	8.42	3%	1%	96%	3.43
1000	50	1	Raw	3.32	4%	96%	0%	0.00
			LogML	4.09	16%	84%	0%	0.00
1000	50	10	Raw	3.27	2%	98%	0%	0.00
			LogML	4.03	16%	83%	1%	0.01
1000	200	1	Raw	8.43	6%	2%	92%	3.46
			LogML	6.89	16%	1%	83%	1.90
1000	200	10	Raw	8.08	8%	5%	87%	3.13
			LogML	7.40	8%	2%	90%	2.42

Scenario 4. Predictor variables $x_1, \dots, x_{20}$ are sampled independently from a Bernoulli(0.2) distribution, predictors $x_{21}, \dots, x_{40}$ are sampled independently from a Bernoulli(0.5) distribution and predictors $x_{41}, \dots, x_{84}$ are sampled from a multivariate normal distribution with mean 0, variance 1 and correlation 0.3. The response is generated according to $y = f(x) + \epsilon$, where $\epsilon \sim N(0, \sigma^2)$ and f is defined as:

$$f(x) = -4 + x_1 + \sin(\pi x_1 x_{44}) - x_{21} + 0.6 x_{41} x_{42} - \exp[-2(x_{42} + 1)^2] - x_{43}^2 + 0.5 x_{44}.$$

Values of $n \in \{500, 1000\}$ and $\sigma^2 \in \{1, 10\}$ were considered when comparing the methods.

Table 4. Variable selection results for Scenario 4. Statistics and notation are as in Table 1.

Selection performance							
n	p	σ^2	Importance	Precision	Recall	F1	r_{miss}
500	50	1	Raw	0.659 (0.02)	0.447 (0.01)	0.523 (0.01)	100%
			LogML	0.925 (0.01)	0.678 (0.01)	0.774 (0.01)	100%
500	50	10	Raw	0.478 (0.03)	0.260 (0.01)	0.322 (0.01)	100%
			LogML	0.615 (0.02)	0.503 (0.02)	0.542 (0.02)	100%
1000	50	1	Raw	0.767 (0.02)	0.508 (0.01)	0.601 (0.01)	100%
			LogML	0.983 (0.01)	0.743 (0.01)	0.841 (0.01)	100%
1000	50	10	Raw	0.614 (0.02)	0.350 (0.01)	0.434 (0.01)	100%
			LogML	0.810 (0.02)	0.598 (0.01)	0.678 (0.01)	100%

Selection size diagnostics (true = 5)								
n	p	σ^2	Importance	Mean	Exact	Under	Over	Mean excess
500	50	1	Raw	4.27	10%	88%	2%	0.03
			LogML	4.43	11%	89%	0%	0.00
500	50	10	Raw	3.62	6%	91%	3%	0.04
			LogML	5.08	18%	68%	14%	0.25
1000	50	1	Raw	4.23	9%	87%	4%	0.06
			LogML	4.55	4%	96%	0%	0.00
1000	50	10	Raw	3.67	5%	94%	1%	0.03
			LogML	4.55	12%	82%	6%	0.06

As can be seen from tables 3 and 4, the log-ML-weighted statistic achieves on average higher scores in every parameter combination of the continuous response/mixed predictors experiments. Within Scenario 3 the largest differences occur for $p = 200$, indicating that the advantages of the weighted statistic may become more pronounced as the number of irrelevant predictors increases.

In Scenario 3 the weighted method outperformed standard VIPs with regard to r_{miss} in every variation of parameters, performing especially well when the total number of predictors p was larger. Looking at the selection size

diagnostics table, we can see that in scenarios where $p = 200$ the average number of selected features was similar between the methods. Thus, log-ML-weighted method's lower r_{miss} percentage does not appear to be explained solely by a larger selected set.

In Scenario 4 both methods missed at least one variable in every repetition. This can partly be attributed to the fact that both methods underestimated the number of relevant features. However, the weighted statistic still achieves substantially higher precision, recall and F1 scores. Even though in some settings the statistic on average selects slightly more variables, the difference in scores cannot be explained entirely by selection size. In particular, for $n = 1000$ and $\sigma^2 = 1$ the log-ML-weighted method underselects more frequently than standard VIPs while its scores remain higher. This suggests that the variables the weighted method selects are more likely to be truly relevant, compared to standard VIPs.

4.4 Mixed Predictors and Binary Response

The last two scenarios consider mixed type predictor variables with a binary response. These scenarios are essentially the classification counterparts of Scenarios 3 and 4: the same functions f are used, but the response is generated through a probit link. In the following definitions Φ denotes the cumulative distribution function of the standard normal distribution.

Scenario 5. Predictor variables $x_1, \dots, x_{\lceil p/2 \rceil}$ are sampled independently from a Bernoulli(0.5) distribution, while predictors $x_{\lceil p/2 \rceil + 1}, \dots, x_p$ are sampled independently from a Uniform(0, 1) distribution. The response y is generated from a Bernoulli($\Phi(f(x))$) where f is defined as in Scenario 3.

Combinations of values $n \in \{500, 1000\}$ and $p \in \{50, 200\}$ were used in comparing the two methods in Scenario 5.

Table 5. Variable selection results for Scenario 5. Statistics and notation are as in Table 1.

Selection performance							
n	p	σ^2	Importance	Precision	Recall	F1	r_{miss}
500	50	1	Raw	0.102 (0.03)	0.032 (0.01)	0.046 (0.01)	100%
			LogML	0.231 (0.03)	0.098 (0.01)	0.132 (0.02)	100%
500	200	1	Raw	0.029 (0.01)	0.032 (0.01)	0.029 (0.01)	100%
			LogML	0.041 (0.01)	0.072 (0.01)	0.052 (0.01)	100%
1000	50	1	Raw	0.067 (0.02)	0.030 (0.01)	0.040 (0.01)	100%
			LogML	0.300 (0.03)	0.132 (0.01)	0.174 (0.02)	100%
1000	200	1	Raw	0.030 (0.01)	0.036 (0.01)	0.031 (0.01)	100%
			LogML	0.048 (0.01)	0.094 (0.01)	0.063 (0.01)	100%

Selection size diagnostics (true = 5)								
n	p	σ^2	Importance	Mean	Exact	Under	Over	Mean excess
500	50	1	Raw	1.41	0%	99%	1%	0.01
			LogML	2.08	2%	98%	0%	0.00
500	200	1	Raw	6.21	12%	24%	64%	1.59
			LogML	9.05	7%	2%	91%	4.08
1000	50	1	Raw	1.52	1%	99%	0%	0.00
			LogML	2.12	5%	95%	0%	0.00
1000	200	1	Raw	6.08	19%	27%	54%	1.58
			LogML	9.72	3%	0%	97%	4.72

Scenario 6. Predictor variables $x_1, \dots, x_{20}$ are sampled independently from a Bernoulli(0.2) distribution, predictors $x_{21}, \dots, x_{40}$ are sampled independently from a Bernoulli(0.5) and predictors $x_{41}, \dots, x_{84}$ are sampled from a multivariate normal distribution with mean 0, variance 1 and correlation 0.3. The response y is generated from a Bernoulli($\Phi(f(x))$) distribution where f is defined as in Scenario 4.

Here we vary only the sample size $n \in \{500, 1000\}$ in comparing the feature selection methods.

Table 6. Variable selection results for Scenario 6. Statistics and notation are as in Table 1.

Selection performance							
n	p	σ^2	Importance	Precision	Recall	F1	r_{miss}
500	50	1	Raw	0.040 (0.01)	0.020 (0.01)	0.026 (0.01)	100%
			LogML	0.088 (0.02)	0.055 (0.01)	0.066 (0.01)	100%
1000	50	1	Raw	0.047 (0.02)	0.018 (0.01)	0.024 (0.01)	100%
			LogML	0.094 (0.02)	0.055 (0.01)	0.067 (0.01)	100%

Selection size diagnostics (true = 6)								
n	p	σ^2	Importance	Mean	Exact	Under	Over	Mean excess
500	50	1	Raw	2.28	1%	99%	0%	0.00
			LogML	3.27	2%	96%	2%	0.05
1000	50	1	Raw	2.25	2%	98%	0%	0.00
			LogML	3.58	7%	90%	3%	0.03

In experiments with a binary response variable both methods performed relatively poorly, as can be seen in tables 5 and 6. Nevertheless, the weighted method achieved higher average scores in both scenarios with every combination of parameters.

Interestingly, there is a larger gain in performance scores in variants of Scenario 5 where $p = 50$ than those where $p = 200$, even though in the latter there was a larger difference in the mean number of selected variables. Therefore, in variants with $p = 50$ where both methods underestimate the number of relevant variables, we can conclude that on average the weighted method’s selection is more often truly relevant. In variants where $p = 200$ the higher precision scores indicate that on average a larger proportion of the variables selected by the weighted method are truly relevant, but the increase in recall could partly be due to larger selected sets.

In Scenario 6 both methods perform poorly in terms of performance scores and both tend to select fewer than the true number of relevant variables. However, the higher scores suggest that the weighted method achieves better partial recovery: like in Scenario 5, higher precision scores indicate that a larger proportion of selected variables are truly relevant, while higher recall

could be in part due to on average larger selected sets.

4.5 Sensitivity to BART Hyperparameters

In addition to the experiments discussed above, an analysis of BART hyperparameter sensitivity was conducted.

To reduce the number of hyperparameter combinations, three hyperparameters were selected for examination: m , β and k . m was chosen because the number of trees in the sum-of-trees controls the amount of redundancy available in the model and, since a smaller number of trees tends to work better for feature selection, values of 10, 20 and 50 trees were tested. β was chosen as it directly influences the tree depth distribution via the tree prior. Values of 1, 2 and 3 were tested, with higher values corresponding to stronger depth penalties. Finally, k was chosen because it controls shrinkage of terminal node parameters and therefore the magnitude of individual tree contributions in the sum-of-trees result. As with β , values of 1, 2 and 3 were chosen, with higher values resulting in greater shrinkage of terminal node parameters towards 0. The other hyperparameters were set as follows: $\alpha = 0.95$

A reference hyperparameter setting with $m = 20$, $\beta = 2$ and $k = 2$ was chosen. Each hyperparameter was then altered one at a time, resulting in seven configurations in total and allowing each hyperparameter’s individual effect to be examined. Interactions between hyperparameters were not examined in this experiment.

Sensitivity was tested using data generated according to Scenarios 1, 4 and 5 as described above. Scenario 1 with $n = 500$ and $p = 50$ was chosen as it is the only scenario where standard VIPs achieved better average F1-scores, even though both methods achieved high average scores. Scenario 4 with $n = 500$ and $\sigma^2 = 10$ was selected as an evaluation on highly noisy data. Additionally, neither method achieved extreme average scores so a different configuration might improve or deteriorate selection performance. Finally, Scenario 5 with $n = 1000$ and $p = 50$ was chosen as a hard binary response problem where neither method achieved strong selection performance in previous experiments.

In addition to precision, recall, F1-score, r_{miss} and the number of selected variables, in this analysis we use Jaccard distance between selected sets

defined as:

$$J(S_0, S_1) = 1 - \frac{|S_0 \cap S_1|}{|S_0 \cup S_1|},$$

where the difference is measured from the set of variables selected by the reference configuration. We also report the difference in the number of true positive features in the selected set ΔTP , as well as the difference in false positive features ΔFP . These statistics allow us to measure both performance sensitivity relative to BART hyperparameters and selection set sensitivity.

Each experiment was repeated 50 times and the reported statistics were averaged over the 50 repetitions. Within each repeat the same generated dataset was used for all seven configurations, while other aspects of model fitting and feature selection (apart from the hyperparameter being varied) were held fixed between the hyperparameter configurations.

Table 7. Hyperparameter sensitivity summary for Scenario 1. The reference configuration is compared with configurations obtained by varying one of m , β , or k while holding the remaining hyperparameters fixed. Reported values are means across replications, with standard errors in parentheses. For each performance metric, the better value within a configuration is shown in bold.

Configuration	Importance	Precision	Recall	F1	r_{miss}	Selected
Reference	Raw	0.95 (0.01)	0.90 (0.02)	0.92 (0.01)	0.46	4.76 (0.12)
	LogML	1.00 (0.00)	0.85 (0.01)	0.91 (0.01)	0.72	4.24 (0.07)
$m = 10$	Raw	0.89 (0.02)	0.78 (0.03)	0.82 (0.02)	0.74	4.40 (0.15)
	LogML	0.98 (0.01)	0.86 (0.02)	0.91 (0.01)	0.64	4.38 (0.09)
$m = 50$	Raw	0.97 (0.01)	1.00 (0.00)	0.99 (0.01)	0.00	5.16 (0.06)
	LogML	1.00 (0.00)	0.95 (0.01)	0.97 (0.01)	0.22	4.76 (0.07)
$\beta = 1$	Raw	0.96 (0.01)	0.95 (0.01)	0.95 (0.01)	0.22	5.04 (0.11)
	LogML	1.00 (0.00)	0.88 (0.02)	0.93 (0.01)	0.56	4.40 (0.08)
$\beta = 3$	Raw	0.95 (0.01)	0.86 (0.02)	0.89 (0.01)	0.58	4.62 (0.15)
	LogML	1.00 (0.00)	0.85 (0.01)	0.92 (0.01)	0.70	4.26 (0.07)
$k = 1$	Raw	0.96 (0.01)	0.91 (0.02)	0.93 (0.01)	0.42	4.78 (0.10)
	LogML	1.00 (0.00)	0.84 (0.02)	0.91 (0.01)	0.74	4.18 (0.08)
$k = 3$	Raw	0.95 (0.01)	0.92 (0.02)	0.93 (0.01)	0.38	4.88 (0.11)
	LogML	1.00 (0.00)	0.87 (0.02)	0.93 (0.01)	0.62	4.34 (0.08)

Table 8. Paired hyperparameter sensitivity relative to the reference configuration for Scenario 1. Panel (a) reports mean signed differences between each altered configuration and the reference; positive values indicate an increase relative to the reference and negative values a decrease. Panel (b) reports mean absolute paired differences. Standard errors are shown in parentheses.

(a) Signed differences								
Configuration	Importance	Δ Prec.	Δ Recall	Δ F1	Δ Sel.	Jaccard dist.	Δ TP	Δ FP
$m = 10$	Raw	−0.06	−0.12	−0.10	−0.36	0.33	−0.60	+0.24
		(0.02)	(0.03)	(0.02)	(0.18)	(0.03)	(0.14)	(0.11)
	LogML	−0.02	+0.01	0.00	+0.14	0.10	+0.06	+0.08
		(0.01)	(0.02)	(0.01)	(0.10)	(0.02)	(0.09)	(0.04)
$m = 50$	Raw	+0.02	+0.10	+0.07	+0.40	0.16	+0.52	−0.12
		(0.01)	(0.02)	(0.01)	(0.13)	(0.02)	(0.09)	(0.09)
	LogML	0.00	+0.10	+0.06	+0.52	0.11	+0.52	0.00
		(0.00)	(0.02)	(0.01)	(0.08)	(0.01)	(0.08)	(0.00)
$\beta = 1$	Raw	+0.01	+0.06	+0.03	+0.28	0.19	+0.28	0.00
		(0.02)	(0.02)	(0.01)	(0.14)	(0.02)	(0.10)	(0.09)
	LogML	0.00	+0.03	+0.02	+0.16	0.09	+0.16	0.00
		(0.00)	(0.02)	(0.01)	(0.09)	(0.02)	(0.09)	(0.00)
$\beta = 3$	Raw	0.00	−0.04	−0.02	−0.14	0.24	−0.18	+0.04
		(0.02)	(0.03)	(0.02)	(0.19)	(0.02)	(0.13)	(0.10)
	LogML	0.00	0.00	0.00	+0.02	0.07	+0.02	0.00
		(0.00)	(0.02)	(0.01)	(0.08)	(0.02)	(0.08)	(0.00)
$k = 1$	Raw	+0.01	+0.02	+0.02	+0.02	0.17	+0.08	−0.06
		(0.02)	(0.02)	(0.01)	(0.15)	(0.02)	(0.11)	(0.09)
	LogML	0.00	−0.01	−0.01	−0.06	0.07	−0.06	0.00
		(0.00)	(0.02)	(0.01)	(0.08)	(0.02)	(0.08)	(0.00)
$k = 3$	Raw	0.00	+0.02	+0.01	+0.12	0.18	+0.10	+0.02
		(0.02)	(0.02)	(0.01)	(0.14)	(0.02)	(0.09)	(0.11)
	LogML	0.00	+0.02	+0.01	+0.10	0.07	+0.10	0.00
		(0.00)	(0.02)	(0.01)	(0.08)	(0.02)	(0.08)	(0.00)

(b) Absolute differences

Configuration	Importance	$ \Delta $ Prec.	$ \Delta $ Recall	$ \Delta $ F1	$ \Delta $ Sel.	Jaccard dist.	$ \Delta $ TP	$ \Delta $ FP
$m = 10$	Raw	0.11	0.17	0.13	1.00	0.33	0.84	0.52
		(0.02)	(0.02)	(0.01)	(0.13)	(0.03)	(0.11)	(0.08)
	LogML	0.02	0.08	0.05	0.42	0.10	0.38	0.08
		(0.01)	(0.01)	(0.01)	(0.08)	(0.02)	(0.07)	(0.04)
$m = 50$	Raw	0.06	0.10	0.08	0.68	0.16	0.52	0.32
		(0.01)	(0.02)	(0.01)	(0.10)	(0.02)	(0.09)	(0.08)
	LogML	0.00	0.11	0.06	0.56	0.11	0.56	0.00
		(0.00)	(0.01)	(0.01)	(0.07)	(0.01)	(0.07)	(0.00)
$\beta = 1$	Raw	0.07	0.10	0.08	0.72	0.19	0.52	0.40
		(0.01)	(0.02)	(0.01)	(0.10)	(0.02)	(0.08)	(0.08)
	LogML	0.00	0.08	0.05	0.40	0.09	0.40	0.00
		(0.00)	(0.01)	(0.01)	(0.07)	(0.02)	(0.07)	(0.00)
$\beta = 3$	Raw	0.08	0.13	0.09	1.06	0.24	0.66	0.44
		(0.01)	(0.02)	(0.01)	(0.11)	(0.02)	(0.10)	(0.08)
	LogML	0.00	0.06	0.03	0.30	0.07	0.30	0.00
		(0.00)	(0.01)	(0.01)	(0.07)	(0.02)	(0.07)	(0.00)
$k = 1$	Raw	0.06	0.09	0.07	0.74	0.17	0.44	0.34
		(0.01)	(0.02)	(0.01)	(0.11)	(0.02)	(0.09)	(0.07)
	LogML	0.00	0.06	0.04	0.30	0.07	0.30	0.00
		(0.00)	(0.01)	(0.01)	(0.07)	(0.02)	(0.07)	(0.00)
$k = 3$	Raw	0.08	0.08	0.07	0.68	0.18	0.38	0.46
		(0.01)	(0.01)	(0.01)	(0.10)	(0.02)	(0.07)	(0.09)
	LogML	0.00	0.06	0.03	0.30	0.07	0.30	0.00
		(0.00)	(0.01)	(0.01)	(0.07)	(0.02)	(0.07)	(0.00)

In Scenario 1 the log-ML-weighted method is consistently less sensitive to the tested hyperparameters than standard VIPs. Across all perturbations the log-ML-weighted method showed smaller mean absolute changes in F1, recall, number of selected variables as well as in Jaccard distance, number of truly relevant features selected and number of false positives.

β and k appear to have a modest effect on average performance. However, in terms of selection sets the log-ML-weighted method appears more stable: Jaccard distances are smaller and there is no change in the number of false positives under the β and k perturbations. The number of detected truly relevant variables varies slightly, but to a lesser extent than for standard VIPs.

Larger sensitivity differences occur for changes in m , particularly for $m = 10$.

The smaller number of trees drastically worsens standard VIPs' average performance, resulting in the only configuration where the log-ML-weighted method outperformed standard VIPs, contrary to the results in the previous section. The $m = 50$ configuration boosts average performance of both methods compared to the reference. In that configuration the log-ML-weighted method on average recovers 0.52 more truly relevant variables, without an increase in false positives. Therefore, among the tested hyperparameters, m is the dominant source of sensitivity for both methods in Scenario 1.

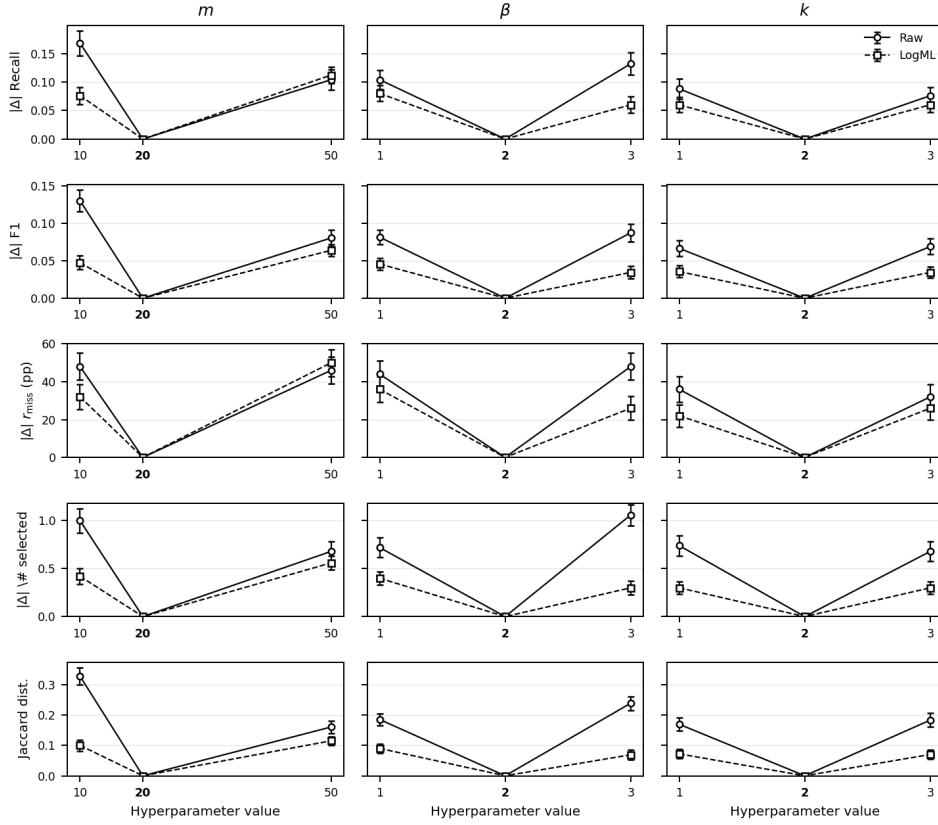


Figure 2. Absolute sensitivity to BART hyperparameters in Scenario 1. Columns correspond to changes in m , β , and k , while rows show the mean absolute change in recall, F1-score, r_{miss} , number of selected variables, and Jaccard distance relative to the reference configuration $(m, \beta, k) = (20, 2, 2)$. Solid and dashed lines represent standard VIPs and log-ML-weighted VIPs, respectively, with error bars showing standard errors.

Table 9. Hyperparameter sensitivity summary for Scenario 4. The reference configuration is compared with configurations obtained by varying one of m , β , or k while holding the remaining hyperparameters fixed. Reported values are means across replications, with standard errors in parentheses. For each performance metric, the better value within a configuration is shown in bold.

Configuration	Importance	Precision	Recall	F1	r_{miss}	Selected
Reference	Raw	0.46 (0.04)	0.24 (0.02)	0.30 (0.02)	1.00	3.68 (0.25)
	LogML	0.60 (0.03)	0.51 (0.03)	0.54 (0.02)	1.00	5.26 (0.21)
$m = 10$	Raw	0.45 (0.03)	0.28 (0.02)	0.33 (0.02)	1.00	3.88 (0.24)
	LogML	0.52 (0.02)	0.52 (0.03)	0.51 (0.02)	0.98	6.08 (0.21)
$m = 50$	Raw	0.55 (0.03)	0.29 (0.02)	0.36 (0.02)	1.00	3.42 (0.21)
	LogML	0.67 (0.03)	0.47 (0.02)	0.54 (0.02)	1.00	4.44 (0.18)
$\beta = 1$	Raw	0.48 (0.03)	0.24 (0.01)	0.31 (0.02)	1.00	3.22 (0.16)
	LogML	0.59 (0.03)	0.43 (0.02)	0.48 (0.02)	1.00	4.64 (0.23)
$\beta = 3$	Raw	0.50 (0.03)	0.30 (0.02)	0.36 (0.02)	1.00	3.92 (0.22)
	LogML	0.56 (0.03)	0.51 (0.03)	0.53 (0.02)	1.00	5.58 (0.22)
$k = 1$	Raw	0.53 (0.03)	0.29 (0.02)	0.37 (0.02)	1.00	3.52 (0.18)
	LogML	0.61 (0.03)	0.48 (0.02)	0.52 (0.02)	1.00	4.90 (0.20)
$k = 3$	Raw	0.49 (0.04)	0.27 (0.02)	0.34 (0.02)	1.00	3.58 (0.21)
	LogML	0.58 (0.03)	0.48 (0.02)	0.51 (0.02)	1.00	5.10 (0.20)

Table 10. Paired hyperparameter sensitivity relative to the reference configuration for Scenario 4. Panel (a) reports mean signed differences between each altered configuration and the reference; positive values indicate an increase relative to the reference and negative values a decrease. Panel (b) reports mean absolute paired differences. Standard errors are shown in parentheses.

(a) Signed differences								
Configuration	Importance	Δ Prec.	Δ Recall	Δ F1	Δ Sel.	Jaccard dist.	Δ TP	Δ FP
$m = 10$	Raw	-0.01	+0.04	+0.04	+0.20	0.77	+0.22	-0.02
		(0.04)	(0.02)	(0.02)	(0.31)	(0.02)	(0.13)	(0.27)
	LogML	-0.07	+0.01	-0.03	+0.82	0.50	+0.04	+0.78
		(0.03)	(0.02)	(0.02)	(0.24)	(0.02)	(0.15)	(0.21)
$m = 50$	Raw	+0.09	+0.05	+0.07	-0.26	0.71	+0.30	-0.56
		(0.04)	(0.02)	(0.02)	(0.33)	(0.03)	(0.12)	(0.28)
	LogML	+0.07	-0.04	0.00	-0.82	0.46	-0.24	-0.58
		(0.03)	(0.02)	(0.02)	(0.23)	(0.03)	(0.10)	(0.21)
$\beta = 1$	Raw	+0.02	0.00	+0.01	-0.46	0.74	0.00	-0.46
		(0.05)	(0.02)	(0.02)	(0.29)	(0.02)	(0.11)	(0.28)
	LogML	-0.01	-0.08	-0.06	-0.62	0.51	-0.48	-0.14
		(0.03)	(0.02)	(0.02)	(0.24)	(0.02)	(0.10)	(0.24)
$\beta = 3$	Raw	+0.04	+0.06	+0.06	+0.24	0.75	+0.36	-0.12
		(0.05)	(0.02)	(0.03)	(0.33)	(0.02)	(0.15)	(0.27)
	LogML	-0.03	0.00	-0.01	+0.32	0.45	0.00	+0.32
		(0.02)	(0.02)	(0.02)	(0.25)	(0.03)	(0.11)	(0.21)
$k = 1$	Raw	+0.07	+0.05	+0.07	-0.16	0.71	+0.32	-0.48
		(0.05)	(0.02)	(0.02)	(0.28)	(0.03)	(0.12)	(0.25)
	LogML	+0.01	-0.04	-0.02	-0.36	0.48	-0.22	-0.14
		(0.03)	(0.02)	(0.02)	(0.24)	(0.02)	(0.12)	(0.20)
$k = 3$	Raw	+0.03	+0.03	+0.04	-0.10	0.73	+0.20	-0.30
		(0.05)	(0.02)	(0.03)	(0.32)	(0.03)	(0.14)	(0.30)
	LogML	-0.02	-0.03	-0.03	-0.16	0.45	-0.20	+0.04
		(0.03)	(0.02)	(0.02)	(0.25)	(0.03)	(0.12)	(0.21)

(b) Absolute differences

Configuration	Importance	$ \Delta $ Prec.	$ \Delta $ Recall	$ \Delta $ F1	$ \Delta $ Sel.	Jaccard dist.	$ \Delta $ TP	$ \Delta $ FP
$m = 10$	Raw	0.25	0.12	0.15	1.80	0.77	0.74	1.46
		(0.03)	(0.01)	(0.01)	(0.18)	(0.02)	(0.09)	(0.17)
	LogML	0.14	0.11	0.11	1.38	0.50	0.64	1.22
		(0.02)	(0.02)	(0.02)	(0.18)	(0.02)	(0.12)	(0.16)
$m = 50$	Raw	0.22	0.10	0.11	1.90	0.71	0.58	1.56
		(0.03)	(0.02)	(0.01)	(0.20)	(0.03)	(0.10)	(0.19)
	LogML	0.16	0.08	0.09	1.30	0.46	0.48	1.18
		(0.02)	(0.01)	(0.01)	(0.18)	(0.03)	(0.08)	(0.14)
$\beta = 1$	Raw	0.25	0.09	0.10	1.54	0.74	0.52	1.38
		(0.03)	(0.01)	(0.01)	(0.21)	(0.02)	(0.08)	(0.20)
	LogML	0.18	0.11	0.12	1.38	0.51	0.64	1.26
		(0.02)	(0.01)	(0.01)	(0.16)	(0.02)	(0.08)	(0.16)
$\beta = 3$	Raw	0.27	0.13	0.16	1.84	0.75	0.80	1.44
		(0.03)	(0.02)	(0.02)	(0.20)	(0.02)	(0.10)	(0.18)
	LogML	0.12	0.09	0.09	1.36	0.45	0.52	1.08
		(0.02)	(0.01)	(0.01)	(0.16)	(0.03)	(0.09)	(0.15)
$k = 1$	Raw	0.26	0.09	0.12	1.56	0.71	0.56	1.40
		(0.03)	(0.02)	(0.02)	(0.17)	(0.03)	(0.10)	(0.17)
	LogML	0.14	0.10	0.10	1.24	0.48	0.62	1.06
		(0.02)	(0.01)	(0.01)	(0.17)	(0.02)	(0.09)	(0.14)
$k = 3$	Raw	0.29	0.11	0.15	1.86	0.73	0.68	1.78
		(0.03)	(0.02)	(0.02)	(0.19)	(0.03)	(0.10)	(0.16)
	LogML	0.15	0.11	0.11	1.24	0.45	0.64	1.04
		(0.02)	(0.01)	(0.01)	(0.17)	(0.03)	(0.09)	(0.15)

In Scenario 4 feature selection is more sensitive to the tested BART hyper-parameters than in Scenario 1, especially in terms of selected sets. Average performance does not vary greatly between the tested configurations and the log-ML-weighted method maintains its advantage with every variation. However, Jaccard distances indicate that selection sets differ substantially from the reference configuration.

Similarly to Scenario 1, β and k have a modest impact on performance, with results for the log-ML-weighted method indicating smaller changes on average compared to standard VIPs. m appears to not have as much of an impact on average performance as in Scenario 1.

Jaccard distances indicate that standard VIPs' selected sets vary greatly compared to the reference with values around 0.75. The log-ML-weighted

method changes selections as well, but to a somewhat lesser extent with Jaccard distances around 0.5. Looking at the true positive and false positive rates, we see that the log-ML-weighted method achieved lower false positive change (in absolute terms) in every configuration and lower true positive change in four out of the six configurations tested, indicating more stability in selected sets. On the other hand, standard VIPs selections on average gained truly relevant variables, while losing false positives. This is reflected in the performance statistics as well, where in every configuration standard VIPs on average improved on the reference.

Overall, the log-ML-weighted method maintained its advantage in performance and exhibited greater selection set stability, whereas standard VIPs were more sensitive to the tested hyperparameters and often improved on the reference configuration.

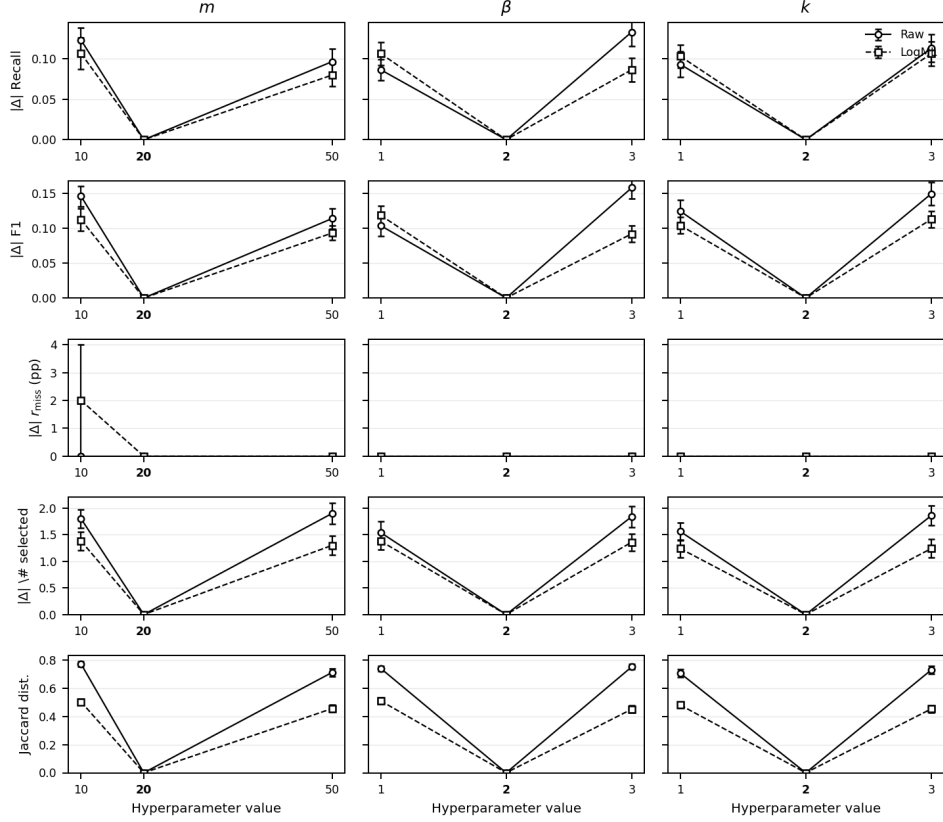


Figure 3. Absolute sensitivity to BART hyperparameters in Scenario 4. Columns correspond to changes in m , β , and k , while rows show the mean absolute change in recall, F1-score, r_{miss} , number of selected variables, and Jaccard distance relative to the reference configuration $(m, \beta, k) = (20, 2, 2)$. Solid and dashed lines represent standard VIPs and log-ML-weighted VIPs, respectively, with error bars showing standard errors.

Table 11. Hyperparameter sensitivity summary for Scenario 5. The reference configuration is compared with configurations obtained by varying one of m , β , or k while holding the remaining hyperparameters fixed. Reported values are means across replications, with standard errors in parentheses. For each performance metric, the better value within a configuration is shown in bold.

Configuration	Importance	Precision	Recall	F1	r_{miss}	Selected
Reference	Raw	0.10 (0.03)	0.05 (0.01)	0.06 (0.02)	1.00	1.74 (0.17)
	LogML	0.29 (0.05)	0.14 (0.02)	0.18 (0.03)	1.00	2.32 (0.18)
$m = 10$	Raw	0.10 (0.04)	0.04 (0.01)	0.05 (0.02)	1.00	1.50 (0.15)
	LogML	0.28 (0.05)	0.12 (0.02)	0.16 (0.03)	1.00	2.06 (0.17)
$m = 50$	Raw	0.15 (0.05)	0.05 (0.01)	0.08 (0.02)	1.00	1.50 (0.14)
	LogML	0.40 (0.05)	0.18 (0.02)	0.23 (0.03)	1.00	1.96 (0.17)
$\beta = 1$	Raw	0.12 (0.04)	0.04 (0.01)	0.06 (0.02)	1.00	1.50 (0.17)
	LogML	0.34 (0.05)	0.16 (0.02)	0.20 (0.03)	1.00	2.42 (0.17)
$\beta = 3$	Raw	0.15 (0.04)	0.05 (0.01)	0.07 (0.02)	1.00	1.68 (0.17)
	LogML	0.31 (0.05)	0.16 (0.03)	0.19 (0.03)	1.00	2.28 (0.21)
$k = 1$	Raw	0.09 (0.03)	0.04 (0.01)	0.05 (0.02)	1.00	1.74 (0.17)
	LogML	0.27 (0.04)	0.14 (0.02)	0.18 (0.03)	1.00	2.18 (0.17)
$k = 3$	Raw	0.06 (0.02)	0.04 (0.01)	0.04 (0.02)	1.00	1.92 (0.17)
	LogML	0.31 (0.04)	0.15 (0.02)	0.20 (0.02)	1.00	2.34 (0.16)

Table 12. Paired hyperparameter sensitivity relative to the reference configuration for Scenario 5. Panel (a) reports mean signed differences between each altered configuration and the reference; positive values indicate an increase relative to the reference and negative values a decrease. Panel (b) reports mean absolute paired differences. Standard errors are shown in parentheses.

(a) Signed differences								
Configuration	Importance	Δ Prec.	Δ Recall	Δ F1	Δ Sel.	Jaccard dist.	Δ TP	Δ FP
$m = 10$	Raw	0.00	-0.01	-0.01	-0.24	0.95	-0.06	-0.18
		(0.04)	(0.02)	(0.02)	(0.21)	(0.02)	(0.08)	(0.19)
	LogML	-0.01	-0.02	-0.02	-0.26	0.93	-0.10	-0.16
		(0.07)	(0.03)	(0.04)	(0.22)	(0.02)	(0.14)	(0.19)
$m = 50$	Raw	+0.05	0.00	+0.01	-0.24	0.98	+0.02	-0.26
		(0.06)	(0.02)	(0.03)	(0.21)	(0.01)	(0.10)	(0.20)
	LogML	+0.11	+0.04	+0.05	-0.36	0.89	+0.18	-0.54
		(0.07)	(0.03)	(0.04)	(0.24)	(0.03)	(0.14)	(0.21)
$\beta = 1$	Raw	+0.02	-0.01	-0.01	-0.24	0.94	-0.04	-0.20
		(0.05)	(0.02)	(0.03)	(0.25)	(0.03)	(0.09)	(0.22)
	LogML	+0.05	+0.02	+0.02	+0.10	0.91	+0.08	+0.02
		(0.06)	(0.03)	(0.03)	(0.26)	(0.03)	(0.14)	(0.25)
$\beta = 3$	Raw	+0.05	0.00	+0.01	-0.06	0.95	0.00	-0.06
		(0.05)	(0.02)	(0.02)	(0.25)	(0.02)	(0.08)	(0.23)
	LogML	+0.02	+0.02	+0.01	-0.04	0.92	+0.08	-0.12
		(0.06)	(0.03)	(0.04)	(0.29)	(0.02)	(0.15)	(0.25)
$k = 1$	Raw	-0.01	-0.01	-0.01	0.00	0.94	-0.06	+0.06
		(0.05)	(0.02)	(0.02)	(0.22)	(0.03)	(0.09)	(0.19)
	LogML	-0.03	0.00	0.00	-0.14	0.90	0.00	-0.14
		(0.06)	(0.03)	(0.03)	(0.26)	(0.03)	(0.14)	(0.21)
$k = 3$	Raw	-0.04	-0.01	-0.02	+0.18	0.95	-0.06	+0.24
		(0.04)	(0.02)	(0.02)	(0.24)	(0.02)	(0.09)	(0.22)
	LogML	+0.02	+0.01	+0.02	+0.02	0.90	+0.06	-0.04
		(0.05)	(0.02)	(0.03)	(0.23)	(0.02)	(0.12)	(0.21)

(b) Absolute differences

Configuration	Importance	$ \Delta $ Prec.	$ \Delta $ Recall	$ \Delta $ F1	$ \Delta $ Sel.	Jaccard dist.	$ \Delta $ TP	$ \Delta $ FP
$m = 10$	Raw	0.17	0.06	0.08	1.16	0.95	0.30	1.02
		(0.04)	(0.01)	(0.02)	(0.14)	(0.02)	(0.07)	(0.13)
	LogML	0.34	0.15	0.20	1.22	0.93	0.74	1.00
		(0.04)	(0.02)	(0.02)	(0.13)	(0.02)	(0.09)	(0.12)
$m = 50$	Raw	0.22	0.08	0.11	1.20	0.98	0.38	1.10
		(0.05)	(0.02)	(0.02)	(0.12)	(0.01)	(0.08)	(0.12)
	LogML	0.34	0.13	0.18	1.28	0.89	0.66	1.26
		(0.05)	(0.02)	(0.03)	(0.17)	(0.03)	(0.11)	(0.14)
$\beta = 1$	Raw	0.20	0.08	0.11	1.36	0.94	0.40	1.24
		(0.04)	(0.02)	(0.02)	(0.16)	(0.03)	(0.08)	(0.14)
	LogML	0.33	0.13	0.17	1.42	0.91	0.64	1.38
		(0.04)	(0.02)	(0.02)	(0.16)	(0.03)	(0.11)	(0.16)
$\beta = 3$	Raw	0.19	0.06	0.09	1.30	0.95	0.32	1.18
		(0.04)	(0.01)	(0.02)	(0.16)	(0.02)	(0.07)	(0.15)
	LogML	0.31	0.14	0.17	1.64	0.92	0.68	1.36
		(0.05)	(0.02)	(0.03)	(0.17)	(0.02)	(0.12)	(0.15)
$k = 1$	Raw	0.18	0.08	0.10	1.20	0.94	0.38	1.06
		(0.04)	(0.01)	(0.02)	(0.13)	(0.03)	(0.07)	(0.12)
	LogML	0.29	0.14	0.17	1.38	0.90	0.68	1.18
		(0.04)	(0.02)	(0.02)	(0.17)	(0.03)	(0.10)	(0.13)
$k = 3$	Raw	0.14	0.07	0.09	1.42	0.95	0.34	1.28
		(0.03)	(0.01)	(0.02)	(0.12)	(0.02)	(0.07)	(0.13)
	LogML	0.29	0.11	0.15	1.22	0.90	0.54	1.08
		(0.03)	(0.02)	(0.02)	(0.15)	(0.02)	(0.09)	(0.14)

As we established in the previous section, Scenario 5 offers an especially hard feature selection problem where both methods perform relatively poorly. However, we can see in the summary table that the log-ML-weighted method obtains better average performance for every hyperparameter combination.

Standard VIPs appear to have greater stability in terms of performance statistics, but this type of analysis is complicated by the fact that the method achieves low performance scores and has little room for deterioration. For example, the average F1-score across all configurations does not exceed 0.07 and therefore there is little room for further deterioration. On the other hand, the log-ML-weighted method achieves F1-scores of ≈ 0.2 , leaving more room for variation between different configurations.

In terms of selection sets it appears that the log-ML-weighted method is

slightly less sensitive to the tested hyperparameters because it achieves lower Jaccard distances than standard VIPs in every configuration. However, the distances are still large (in the range (0.89, 0.93) for the log-ML-weighted method, compared to (0.94, 0.98) for standard VIPs). On the other hand, it should be noted that both methods underselect, with the mean number of selected variables around 2, so even a change of one variable can produce a large Jaccard distance.

These results demonstrate that the conclusions drawn in sections 4.2 - 4.4 are robust to the tested changes in BART hyperparameters m , β and k . In Scenario 1, both methods remain competitive, with standard VIPs slightly outperforming the log-ML-weighted method in almost all configurations. The log-ML-weighted method maintains its performance advantage in Scenario 4, with its lowest F1-score across all configurations (0.48) outperforming standard VIPs' highest (0.37). In Scenario 5 both methods perform poorly, but the log-ML-weighted method achieves better results in all configurations.

In Scenarios 1 and 4, where relative sensitivity can be meaningfully compared, log-ML-weighted feature selection appears less sensitive to changes in tested BART hyperparameters both in terms of performance statistics and selection sets. Scenario 5 does not offer a fair comparison of relative stability. However, if we look at the mean absolute differences of performance statistics for the log-ML-weighted method, we can see that they are not unusually large compared to Scenario 4 considering the difficulty of the binary-response problem.

5 Conclusion

In this report we offer a weighting scheme meant to build upon variable inclusion proportions by adding tree structure information to the variable importance statistic. The goal of this approach was to improve feature selection performance while preserving the relative structure of VIPs which enables us to use existing permutation based feature selection schemes.

In the six simulation experiments the log-ML-weighted method generally achieved better precision, recall and F1-scores, with the clearest gains in the high dimensional continuous-response scenarios. One exception to this was Scenario 1 with $p = 50$ where standard VIPs achieved better scores on average because the log-ML-weighted method tended to underselect. The binary-

response scenarios were difficult for both methods, but the log-ML-weighted method achieved better partial recovery.

The sensitivity analysis confirmed that the main performance conclusions persist under the tested changes in BART hyperparameters m , β and k . Results for Scenarios 1 and 4 indicate that the log-ML-weighted method might be less sensitive to changes in the tested hyperparameters, while the same conclusion could not be drawn in Scenario 5 where standard VIPs reached a performance floor. However, Scenario 5 demonstrated how stability in performance statistics is not equivalent to selection-set stability.

The main drawback to the proposed log-ML-weighted approach is the fact that, due to the subtree collapsing scheme, weights between nodes overlap. This means a split might obtain a higher weight due to subsequent splits in the subtree, which could favour splits closer to the root of the tree.

This report compared the two methods on only six synthetic scenarios, with no real data examples. Similarly, only three BART hyperparameters were chosen to test hyperparameter sensitivity. These limitations offer directions for further research: comparisons on broader simulation families and real datasets, as well as sensitivity to other BART hyperparameters and effects of hyperparameter interactions. Nevertheless, the results in this report show the log-ML-weighted importance statistic is a promising addition to standard VIPs which improves relevant variable recovery in many tested settings.

References

- [1] H. A. Chipman, E. I. George, and R. E. McCulloch. “BART: Bayesian additive regression trees”. In: *The Annals of Applied Statistics* 4.1 (Mar. 1, 2010). DOI: 10.1214/09-A0AS285.
- [2] J. Bleich et al. “Variable selection for BART: An application to gene regulation”. In: *The Annals of Applied Statistics* 8.3 (Sept. 1, 2014). DOI: 10.1214/14-A0AS755.
- [3] C. Luo and M. J. Daniels. *Variable Selection Using Bayesian Additive Regression Trees*. Dec. 28, 2021. DOI: 10.48550/arXiv.2112.13998.
- [4] H. Chipman et al. “Bayesian Additive Regression Trees, Computational Approaches”. In: *Wiley StatsRef: Statistics Reference Online*. Ed. by R. S. Kenett et al. 1st ed. Wiley, Feb. 15, 2022, pp. 1–23. DOI: 10.1002/9781118445112.stat08288.

- [5] J. Hill, A. Linero, and J. Murray. “Bayesian Additive Regression Trees: A Review and Look Forward”. In: *Annual Review of Statistics and Its Application* 7.1 (Mar. 9, 2020), pp. 251–278. DOI: 10.1146/annurev-statistics-031219-041110.
- [6] T. Gneiting and A. E. Raftery. “Strictly Proper Scoring Rules, Prediction, and Estimation”. In: *Journal of the American Statistical Association* 102.477 (Mar. 2007), pp. 359–378. DOI: 10.1198/016214506000001437.
- [7] Y. V. Tan and J. Roy. “Bayesian additive regression trees and the General BART model”. In: *Statistics in Medicine* 38.25 (Nov. 10, 2019), pp. 5048–5069. DOI: 10.1002/sim.8347.
- [8] A. Horiguchi, M. T. Pratola, and T. J. Santner. *Assessing variable activity for Bayesian regression trees*. Sept. 15, 2020. DOI: 10.48550/arXiv.2005.13622.